4.3 Centralized configuration management with Spring Cloud Config Server

With environment variables, you can externalize your application's configuration and follow the 15-Factor methodology. However, there are some issues they cannot handle:

- Configuration data is as important as the application code, so it should be handled with the same care and attention, starting from its persistence. Where should you store configuration data?
- Environment variables don't provide granular access control features. How can you control access to configuration data?
- Configuration data will evolve and require changes, just like application code. How should you keep track of the revisions to configuration data? How should you audit the configuration used in a release?
- After changing your configuration data, how can you make your application read it at runtime without requiring a full restart?
- When the number of application instances increases, it can be challenging to handle configuration in a distributed fashion for each instance. How can you overcome such challenges?
- Neither Spring Boot properties nor environment variables support configuration encryption, so you can't safely store passwords. How should you manage secrets?

The Spring ecosystem offers many options to address those issues. We can categorize them into three groups.